

Sparsh Jain

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EDUCATION

Rutgers University

Master of Science in Data Science, GPA: 3.9/4.0

May 2024

New Brunswick, NJ

- Relevant Courses: Data Mining; NLP; Statistical Modelling; Database Management; Regression & Time Series
- Attained pinnacle in Merck's challenge ([Interpretability in Machine Learning models](#))

Maulana Azad National Institute of Technology

Bachelor of Technology in Electronics & Communication Engineering, GPA: 8.45/10

May 2022

Bhopal, India

SKILLS

Programming & Development: Python, SQL, R, PostgreSQL, PyTorch, TensorFlow, C++, Flask, PySpark, Apache Spark

Big Data & Cloud Platforms: MongoDB, Snowflake, Azure Databricks, Google Cloud Platform, AWS Sagemaker, Docker, Kubernetes

Data Analysis & Visualization: Matplotlib, Power BI, Tableau, MS Excel, SAS

Machine Learning & AI: Transformers, LLMs, Langchain, Gen AI, MLOps, Llama, A/B Testing

Certifications: Machine Learning & Statistics, Stanford University; Data Science Specialization, IBM; SQL, UC Davis

EXPERIENCE

Generative AI Engineer

Aug 2024 - Present

Crowd doing || *Python, Transformers, SQL, Llama, Langchain*

Remote

- Implementing data preprocessing pipelines, and building a database that stores over 10K+ summary entries using SQL database
- Fine-tuning a language model to generate concise summaries for eco-friendly organizations, enhancing content efficiency by 30%

Data Scientist

Jun 2023 - May 2024

Health Galaxy (MIT) || *Python, R, MS Excel, GCP, Scikit-Learn, Tableau, MLOps*

Remote

- Engineered and deployed a Bayesian model for heart disease risk prediction by processing 150K datasets. Also, spear-headed MLOps practice and utilized CI/CD pipelines for automated model training and validation, reducing deployment time by 60%
- Designed interactive Tableau dashboards to monitor patient health metrics, boosting efficiency in tracking key indicators by 30%
- Initiated A/B testing to optimize dashboard layout and enhance user experience. Laid the groundwork for inference scaling of predictive models to impact 50,000+ patients annually using Google Cloud Platform's autoscaling feature

Research Engineer Intern

Jan 2022 - Jun 2022

Samsung R&D Institute || *Python, SQL, Android Studio, Kotlin*

Noida, India

- Utilized a data extraction pipeline using Python to collect Mobile Network Code (MNC) & Mobile Country Code (MCC) data from devices, storing it in an SQL database for efficient retrieval and enabling seamless integration of IMS features into the WIFI calling system
- Engineered an Android application and server system that reduced WIFI calling entitlement processing time by 99.96% (from 0.9 to 0.0004 seconds) for 50M+ US users, resulting in a 30% increase in WIFI calling adoption and, a 25% improvement in call quality ratings

Data Scientist Intern

Jul 2021 - Oct 2021

KPMG || *Python, R, MS Excel, Azure DB, PySpark, Scikit-Learn*

Gurgaon, India

- Developed Python scripts to process and analyze historical sales data from over 65K retail outlets, integrating geospatial and time-series data from various public sources to create a comprehensive Retail Network Planning Tool for a major Oil Company
- Optimized and deployed models in Databricks using Python through XGBoost, enhancing prediction accuracy by 30% for retail outlet performance and potential new locations, projected to increase in annual revenue by approximately \$5M in the following 3 years

Data Analyst Intern

Mar 2021 - May 2021

Balco Ltd || *Python, Selenium, MS Excel, SQL, Tableau*

Korba, India

- Constructed a Python-based web scraping tool using Selenium to extract time-dependent power sale data from the Govt. Power Ministry website, automating data collection for over 10K daily transactions and reducing manual data entry time by 85%
- Leveraged Tableau to perform a 3-year projection of the power bidding budget by analyzing historical trends across 5+ years of data

PROJECTS

Squad 2.0 || *Python, PyTorch, NLP, Transformers, Neural Networks* ([GitHub](#))

Dec 2023

- Implemented a BERT-based Question Answering model using PyTorch on the SQuAD 2.0 dataset, processing 40K question-answer pairs and achieving a significant 40% improvement over the baseline in model performance for irrelevant questions
- Optimized the model with a 12-layer transformer architecture and a custom loss function to handle 512-character contexts and ensure accurate detection of irrelevant questions, achieving final metrics of 42% F1-score and 36% Exact Match on 5K test questions

Twitter Search Engine || *Python, Flask, JSON, SQL, Mongo DB, Natural Language Processing(NLP)* ([GitHub](#))

Apr 2023

- Engineered a high-performance Twitter search engine using Python and Flask, processing over 100K tweets with NLP techniques including TF-IDF vectorization and cosine similarity for content-based ranking, resulting in a 40% improvement in search relevance
- Implemented an LRU caching mechanism that reduced query response times from seconds to 0.00009 seconds, a 10^5 x performance boost, while leveraging both SQL database and MongoDB for efficient data storage, retrieval, and complex query handling

IMDb Rating Analysis || *R, Regression* ([GitHub](#))

Dec 2022

- Analyzed 5,855 IMDb movie ratings using R, revealing a 0.5-point increase in average ratings from 1990-2020, and built a linear regression model to predict the movie ratings for recent years, achieving an R-squared value of 0.81
- Created 15+ visualizations with ggplot, uncovering trends like a 300% increase in movie releases from 1980-2020, and conducted hypothesis tests that showed statistically significant differences across genres, directors, and duration of the movie

License Plate Recognition || *Python, TensorFlow, Object Detection, Open CV, CNN, Flask*; Published in [IEEE](#)

Aug 2021

- Developed an advanced License Plate Recognition System using Inception and Single Shot Detector neural networks to achieve 94.36% prediction accuracy on 10K images; implemented innovative text segmentation techniques that improved accuracy from 91.6% to 92.75%